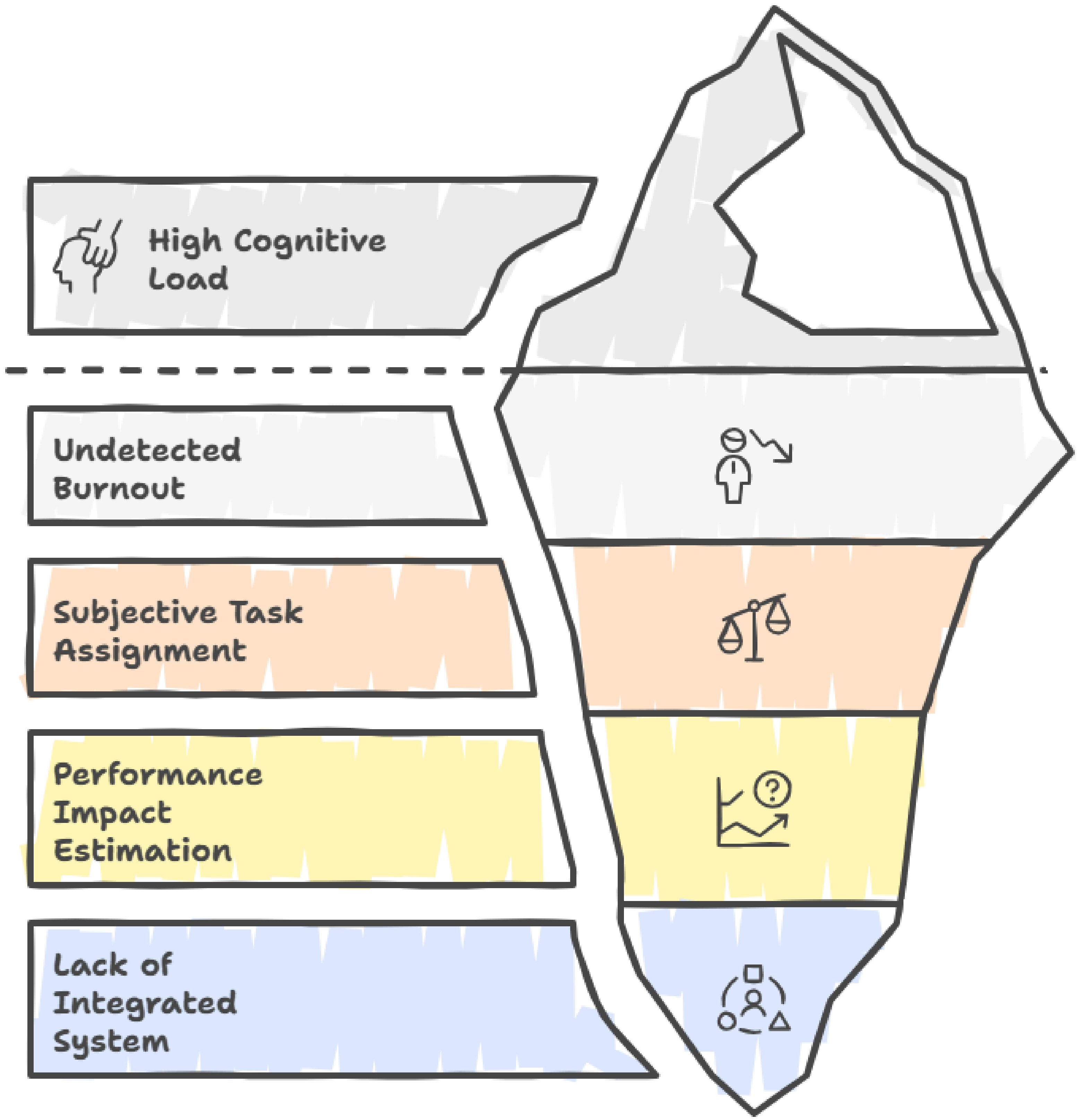
NeuroHR: AI-Driven Cognitive Load & Workforce Performance Estimator

An ML-Based Framework for Burnout Prediction, Performance Impact Estimation, and HR Decision Support

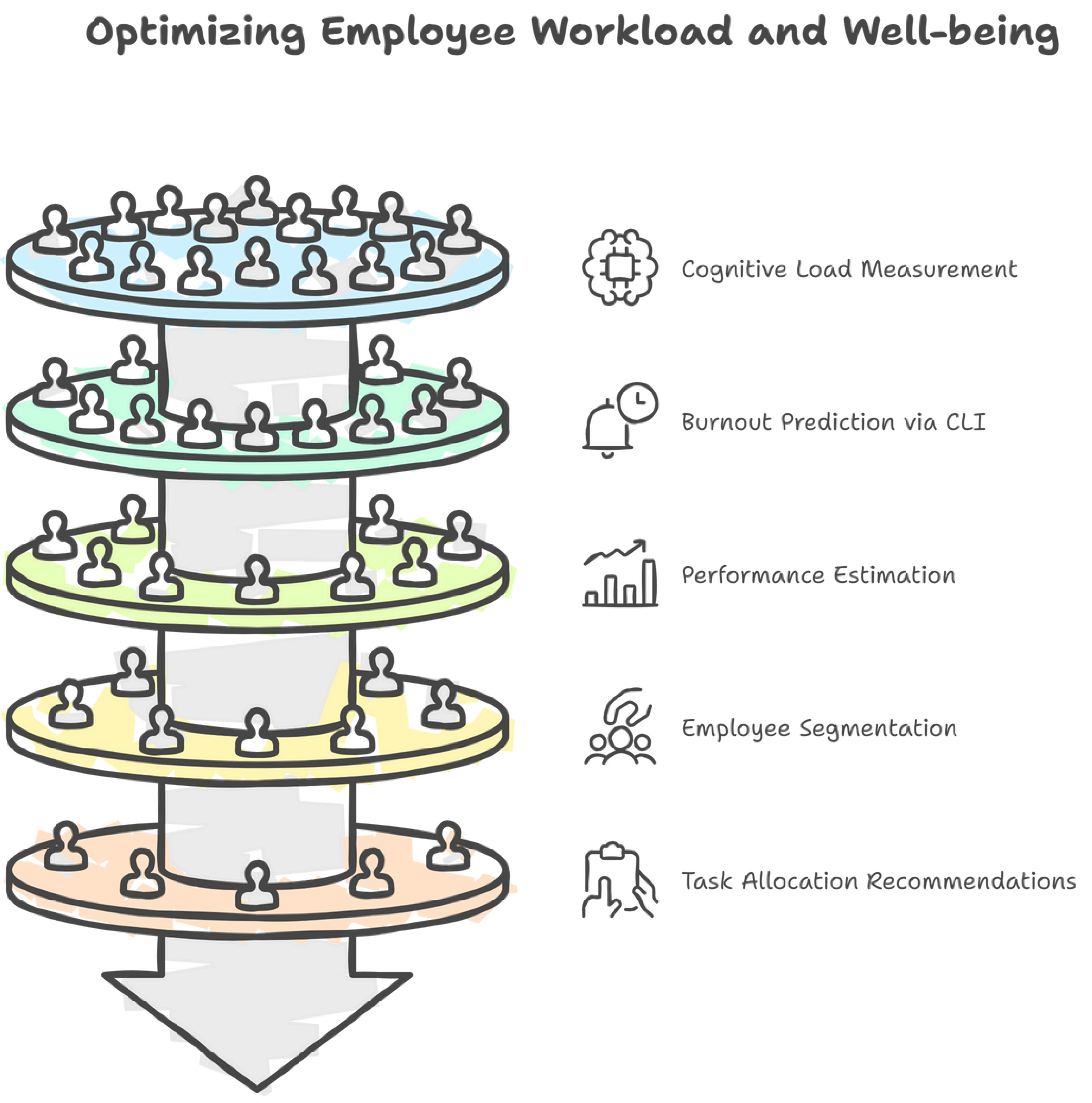
Problem Statement

- Modern workplaces create **high cognitive load** due to **constant multitasking, meetings, and deadlines.**
- **Burnout often goes undetected**, as HR lacks objective and real-time measurement of mental workload.
- **Task** assignment is largely subjective, leading to **overload** of certain employees.
- Organizations **cannot estimate how new tasks will affect performance** or well-being.
- No integrated system exists to use HR + behavioral data for **early burnout prediction** and workload optimization.



Proposed Solution

- Build an AI system that can quantitatively **measure employee cognitive load** using available HR and workload data.
- Develop predictive models that can identify **early signs of burnout** before they appear in performance reviews.
- Estimate the **expected performance change if additional tasks** are assigned to an employee.
- Group employees into **workload and stress segments**, helping HR understand different risk profiles.
- Provide HR with data-driven recommendations for **task allocation who can take more work, who needs support, and who must not be overloaded.**



What is Cognitive Load Index (CLI)?

- Cognitive Load Index (CLI) is a numerical score (0–1) that quantifies an **employee's mental workload and stress level.**
- It combines multiple factors such as **workload hours, task switching, after-hours work, stress levels, sentiment, and focus time.**
- A higher CLI indicates **higher mental strain**, **lower capacity**, and greater **burnout risk.**
- CLI helps HR identify employees who are **overloaded, near burnout, or capable of taking new tasks.**
- It serves as the **core metric used in prediction, clustering, and task-assignment decisions a**cross the entire system.

CLI Mathematical Modeling:

cli = (0.20*wc_norm + 0.12*(1-focus_norm) + 0.10*switch_norm + 0.10*(1-sentiment_norm) + 0.15*stress_norm + 0.08*after_hours_norm + 0.05*(1-salary_exp_norm) + 0.08*overdue_norm + 0.06*vacation_deficit_norm + 0.06*response_time_norm + noise)

The equation adds weighted signals of workload, stress, sentiment, focus, overtime, fairness, and responsiveness to compute a normalized cognitive load score between 0 and 1.

Dataset Preprocessing & Validation

Step 1: Data Cleaning & Validation

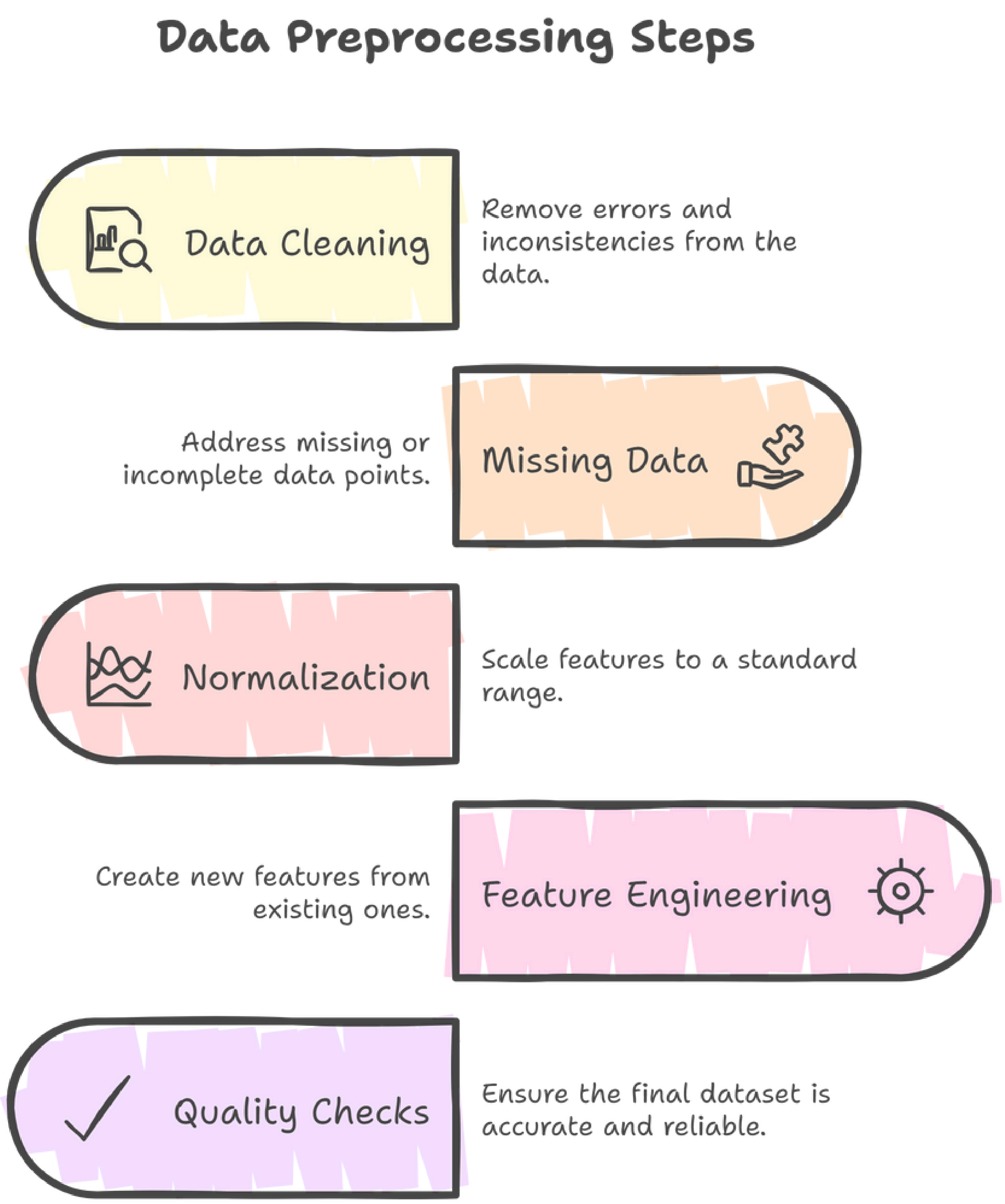
- Removed invalid values and corrected inconsistencies (e.g., tenure, salary, workload ranges).
- Ensured all numeric features fall within realistic corporate limits.

Step 2: Handling Missing & Noisy Data

- Filled missing numeric values using median; categorical using mode.
- Smoothed noisy behavioral scores (stress, sentiment, response time).

Step 3: Normalization of Key Behavioral Features

- Converted workload, stress, sentiment, focus, overtime, and response-time signals into normalized (0–1) scales.



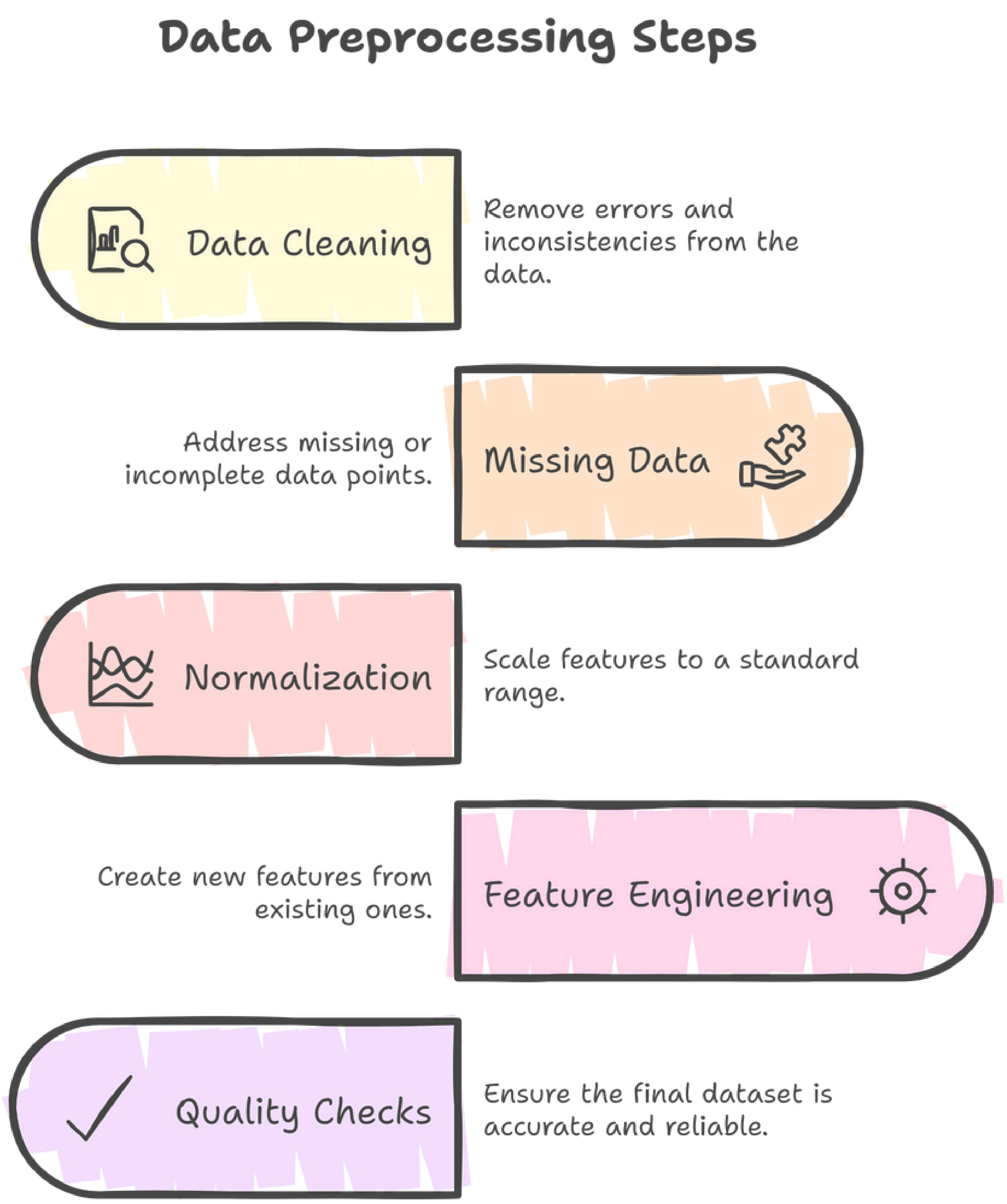
Dataset Preprocessing & Validation

Step 4: Feature Engineering Added to Base HR Dataset

- CLI_v2 (computed cognitive load index using weighted formula).
- ΔPerformance (Delta_if_Assign): expected performance drop for assigning new tasks.
- Delta_no_Assign, Risk_Category, Recommendation, Attrition_Label, Workload Segments, Performance Segments.
- Enhanced HR dataset into a Cognitive-Aware Employee Profile.

Step 5: Final Dataset Quality Checks

- Ensured zero missing values, valid feature ranges, no duplicates, consistent distributions, and clipped outliers.
- Produced a fully cleaned, feature-rich dataset suitable for regression models, clustering, and simulation-based decision workflows.



Model 1: CLI Prediction (RF Regression)

Goal of Model 1: Predict the Cognitive Load Index (CLI_v2) using behavioral, workload, and sentiment-based features.

Step 1: Model Choice: Random Forest Regressor

- Selected due to its ability to capture non-linear relationships and feature interactions.
- Handles mixed numeric + categorical data well.
- More robust to outliers compared to linear models.

Step 2: Key Hyperparameters Used

- n_estimators = 200 //number of trees; more trees improve accuracy and stability.
- max_depth = 15 //limits tree depth to prevent overfitting.
- min_samples_split = 10 //minimum samples needed to split a node, reducing noisy splits.
- min_samples_leaf = 6 //ensures each leaf has enough samples for smoother learning.
- random_state = 42 //fixes randomness for reproducible results.
- max_features= 'sqrt' //Uses √features per split to reduces variance
- n_jobs= -1 //Parallel training for faster compute

(Chosen to balance model accuracy and overfitting control.)

Model 1: CLI Prediction (RF Regression)

Step 3: Training & Validation

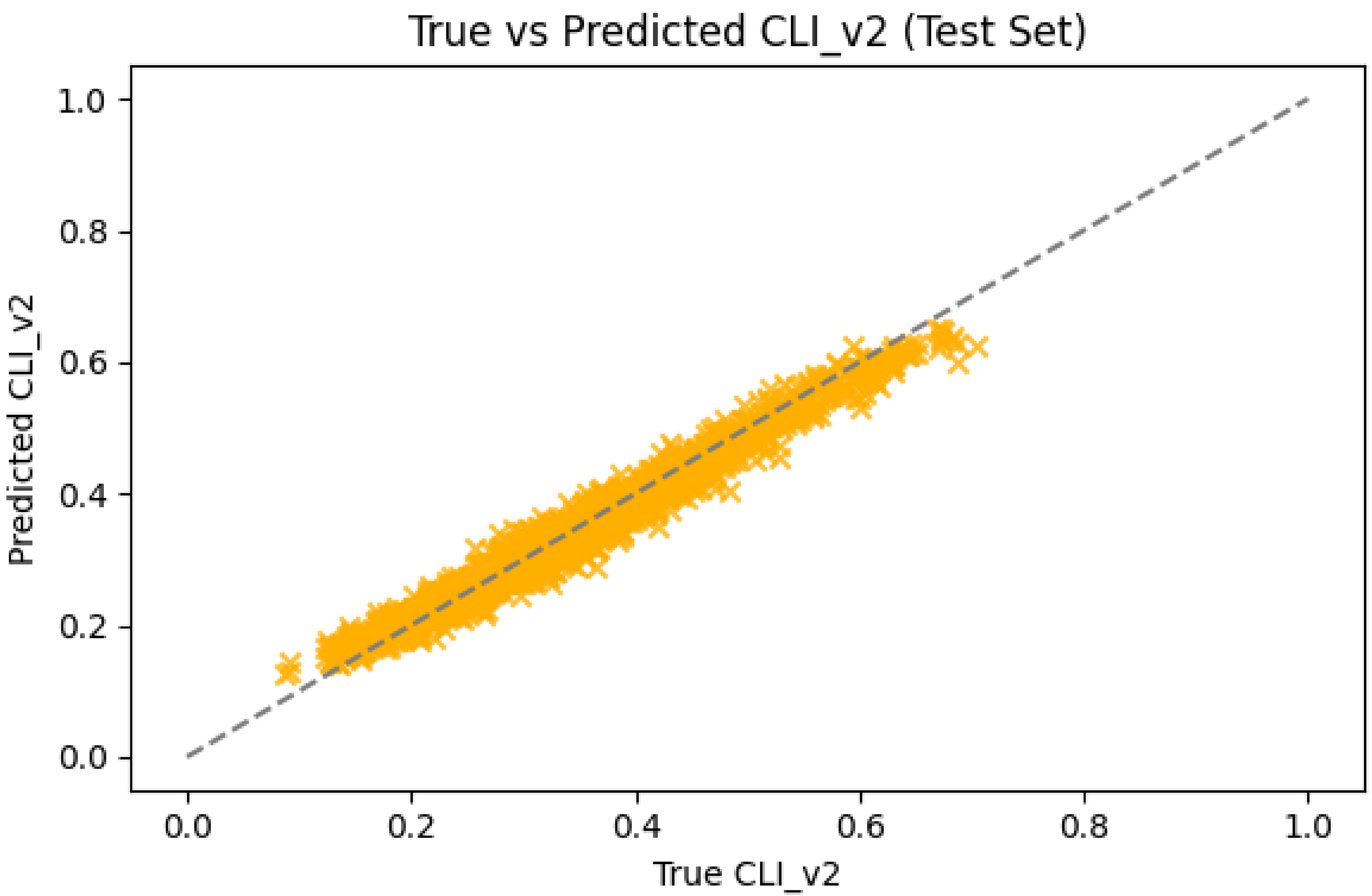
- Train/Test split: 80% / 20%
- Applied 5-fold cross-validation to ensure model stability.
- Used ColumnTransformer + OneHotEncoder for categorical fields (Role Level, Department, Project Phase).

Step 4: Performance Metrics

- MAE (Mean Absolute Error) //evaluates average prediction error.
- RMSE //penalizes larger prediction mistakes.
- R^2 Score //measures model's explanatory power.
- Achieved high R^2 and low error, indicating strong predictive accuracy.

Step 5: R^2 Gap & 5-Fold Cross-Validation Stability

- Compared Train R^2 vs Test R^2 to check for overfitting like the gap was minimal, indicating strong generalization.
- Performed 5-Fold Cross-Validation, and the R^2 scores across all 5 folds were highly consistent, confirming that the model is stable and not sensitive to train/test splits.



True vs Predicted CLI_v2 (Test Set)
Predicted CLI_v2
True CLI_v2
0.0
0.2
0.4
0.6
0.8
1.0

Model 2: Predicting Performance Drop (RFR)

Goal of Model 2: Predict the expected change in performance (ΔPerformance) when a new task is assigned, using cognitive load, workload strain, capacity, stress, and recent performance trends.

Step 1: Model Choice: Random Forest Regressor

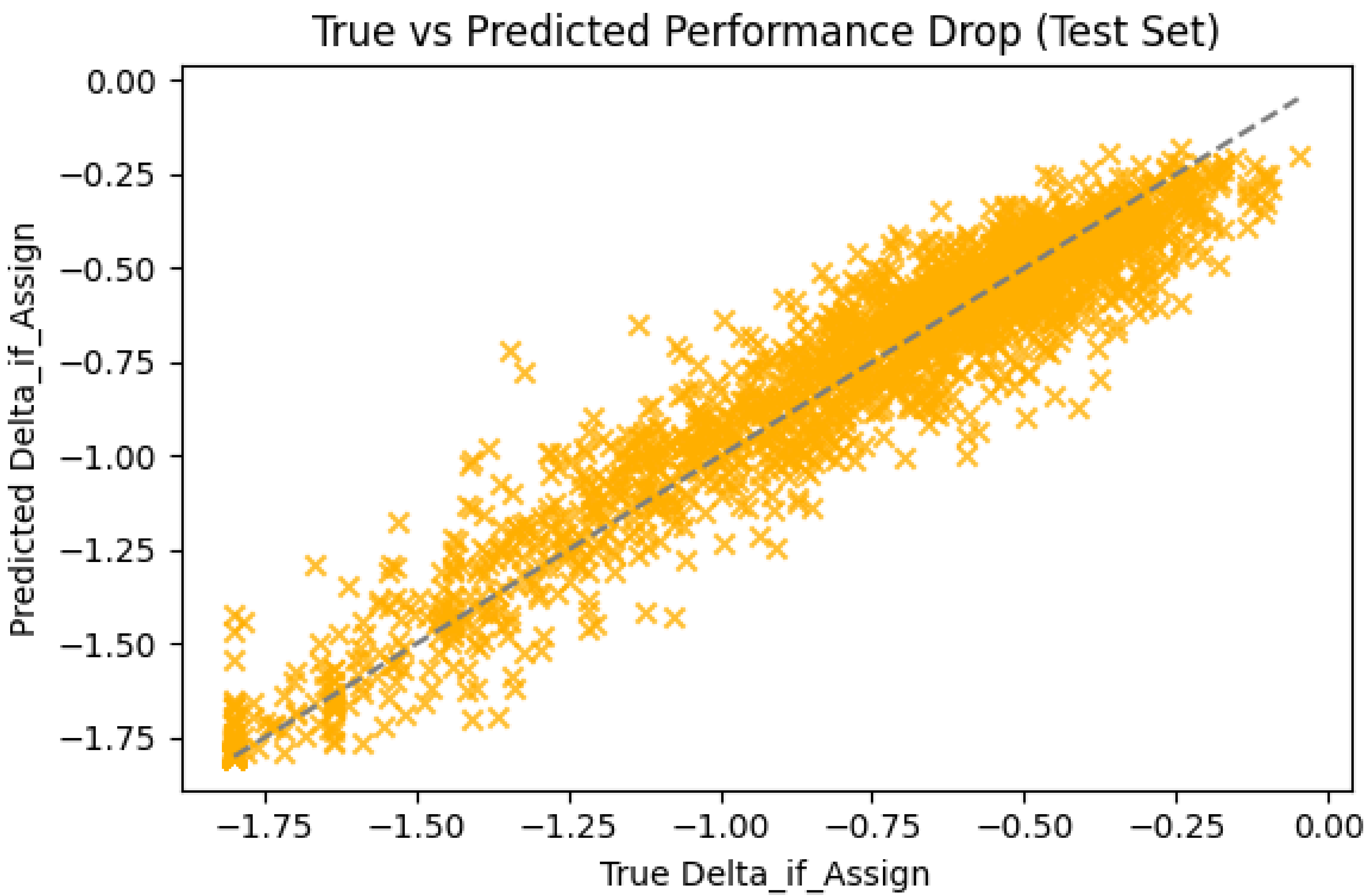
- Handles non-linear relationships between workload + stress + performance.
- Robust to noise and mixed HR behavioral data.
- Suitable for sensitive target (performance drop).

Step 2: Key Hyperparameters

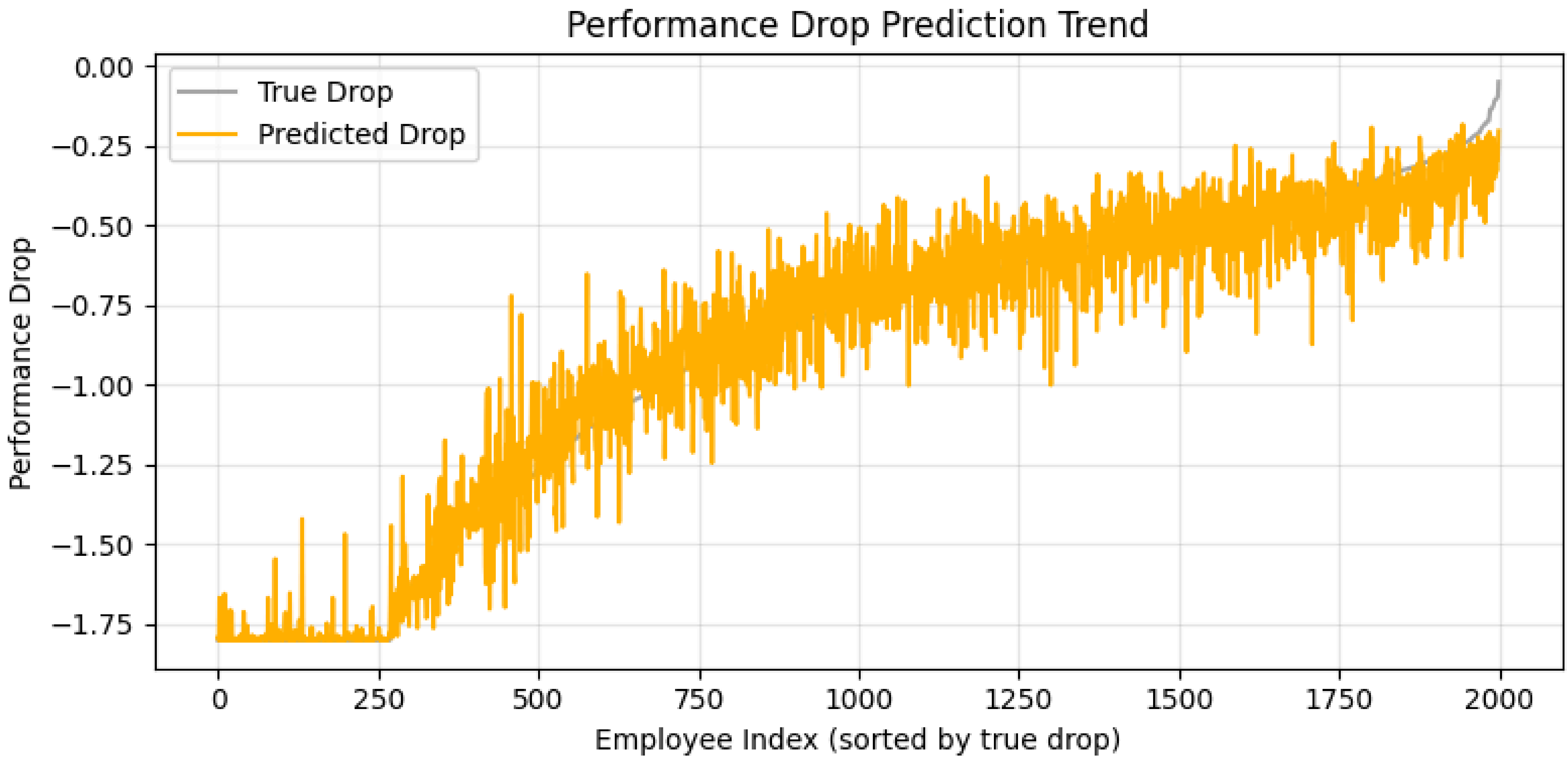
- n_estimators= 200 //Stable ensemble predictions
- min_samples_leaf= 6 //Prevents overfitting on noisy performance signals
- random_state= 42 //Reproducibility

(No max_depth or split constraints as model kept simpler to avoid overfitting)

Step 3, 4 and 5: Training & Validation, Evaluation Metrics and R^2 Gap & Cross-Validation check (Same process as Model 1)



True vs Predicted Performance Drop (Test Set)
Predicted Delta_if_Assign
True Delta_if_Assign
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75



Performance Drop Prediction Trend
True Drop
Predicted Drop
Performance Drop
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
0
250
500
750
1000
1250
1500
1750
2000
Employee Index (sorted by true drop)

Model 3: : Unsupervised Clustering (K-Means) + PCA for visualization and Employee Segmentation

Goal of Clustering Model 3: Identify distinct cognitive like behavioral employee groups (segments) using unsupervised learning to support HR in workload planning, burnout prevention, and role-based task allocation.

Step 1: Method: K-Means Clustering (k=4)

- Groups employees based on similarity in cognitive load, stress, workload, sentiment, and wellbeing.
- Helps HR understand patterns, not predictions e.g., who is stable, who is high stress, etc.
- Works well because variables were standardized and highly behavioral.

Step 2: Features Used for Clustering

The following 11 quantitative behavioral attributes were used: CLI_v2, Self_Reported_Stress, Weekly_Workload_Hours, Workload_to_Capacity_Ratio, Task_Switch_Frequency, Focus_Time_Ratio, Overtime_Hours_30d, After_Hours_Activity_Hours, Sentiment_Score, Survey_Wellbeing_Score, Language_Tone_Stress_Index
These capture mental load, workload burden, behavioral strain, emotional tone, and wellbeing.

Model 3: : Unsupervised Clustering (K-Means) + PCA for visualization and Employee Segmentation

Step 3: Preprocessing

(Same preprocessing approach logic as previous models)

- StandardScaler → ensures all features contribute equally
- No categorical encoding needed
- Dataset transformed to normalized numerical space
- PCA used for 2D visualization only (not for clustering)

Step 4: Clustering Process

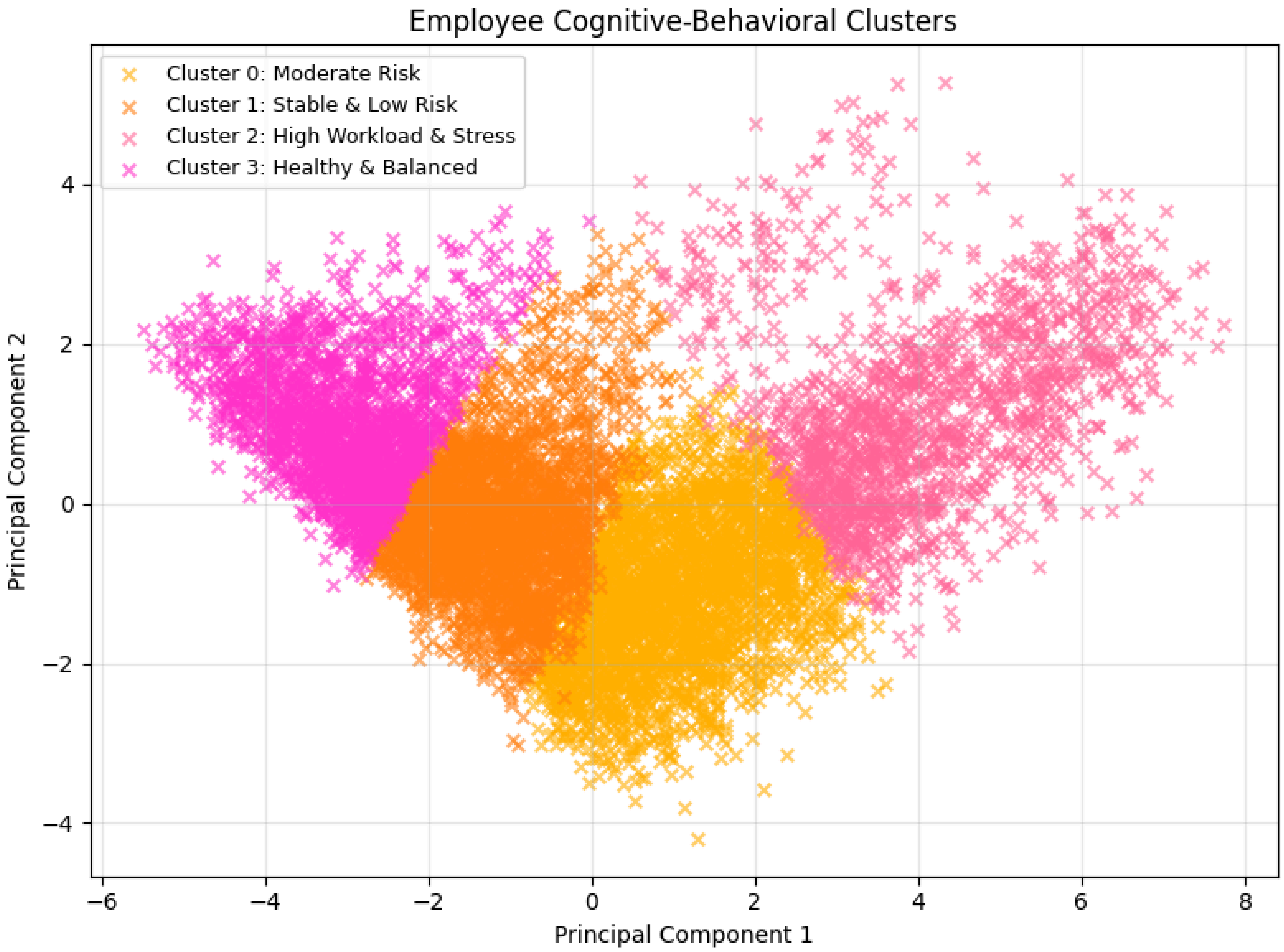
- k = 4 chosen based on:
  - Separation in PCA space
  - Clear behavioral group boundaries
  - HR interpretability
- For each cluster, average values of CLI, Stress, and Workload Ratio computed
- Custom logic added to label clusters as meaningful HR categories

Model 3: : Unsupervised Clustering (K-Means) + PCA for visualization and Employee Segmentation

Step 5: Cluster Profiling & Interpretation
After clustering, each group was analyzed using its average CLI, stress level, workload ratio, and behavioral patterns. The four clusters naturally aligned into meaningful HR personas:

- **Cluster 0:** Healthy & Balanced //Low CLI, low stress, good wellbeing, stable workload; employees are well-managed and can take moderate additional tasks.
- **Cluster 1:** Stable & Low Risk //Slightly higher workload but manageable stress; efficient workers with good focus time; reliable for regular assignments.
- **Cluster 2:** Moderate Risk //Increasing workload pressure, rising stress, reduced focus; early indicators of fatigue; requires monitoring and controlled task allocation.
- **Cluster 3:** High Workload & Stress //Elevated CLI, high stress, frequent overtime and task switching; strong indicators of burnout risk; HR should avoid assigning new tasks.

These cluster personas help HR understand workforce health, identify at-risk groups, and design targeted interventions.



Employee Cognitive-Behavioral Clusters
Cluster 0: Moderate Risk
Cluster 1: Stable & Low Risk
Cluster 2: High Workload & Stress
Cluster 3: Healthy & Balanced
Principal Component 1
Principal Component 2
−6
−4
−2
0
2
4
6
8
−4
−2
0
2
4

Final Integrated Simulation System (CLI + Burnout Probability + ΔPerformance + Clustering)

Goal of Simulation Module: Provide a **real-time decision support system** that predicts an employee's **cognitive load**, **expected performance drop** if assigned new work, assigns a **risk category**, and identifies the **employee's cluster persona** like all combined to generate an HR-ready **workload recommendation**. To demonstrate correctness, the simulation **evaluates four different employee scenarios**, each representing a unique workload–stress profile.

Step 1: Inputs to the Simulation Engine
The simulation uses a minimal set of high-impact behavioral and workload features (user-entered or system-collected), including:

- Weekly workload hours, overtime, after-hours work
- Stress level, sentiment score, focus time ratio
- Task switching frequency, active task count
- Performance trend slope and current performance
- Salary-to-experience ratio

+ Basic categorical attributes (Role level, Department). These features collectively describe mental load, cognitive strain, work behavior, and recent performance signals.

Final Integrated Simulation System (CLI + Burnout Probability + ΔPerformance + Clustering)

Step 2: Model 1 Predict Cognitive Load (CLI_v2)

- Uses Random Forest Regression Model 1 loaded from cli_model.pkl
- (pickle.load() is used; Pickle is Python's tool for saving/loading trained ML models.)
- Generates a continuous CLI score (0–1)
- Converts CLI into a Burnout Risk Category: Low / Medium / High / Critical
- Used as the primary indicator of mental strain

Burnout Probability Formula (exact used in code):

Burnout_Prob = 1 / (1 + exp(-10 * (CLI - 0.55)))

This logistic transformation converts CLI into a smooth probability curve.

Step 3: Model 2 Predict Performance Drop (ΔPerformance)

- Uses Random Forest Regression Model 2 loaded from delta_model.pkl
- Estimates how much performance will decline if a new task is assigned
- Negative ΔPerformance indicates expected degradation
- Helps quantify task shock sensitivity

Final Integrated Simulation System (CLI + Burnout Probability + ΔPerformance + Clustering)

Step 4: Model 3 Assign Behavioral Cluster (K-Means)

- Uses the employee's predicted CLI + workload + stress features
- Loaded from clustering_artifacts.pkl (contains K-Means model, PCA, scaler, and cluster descriptions)
- Maps employee into one of four clusters:
- Healthy, Stable, Moderate Risk, High Workload & Stress
- Provides long-term behavioral interpretation

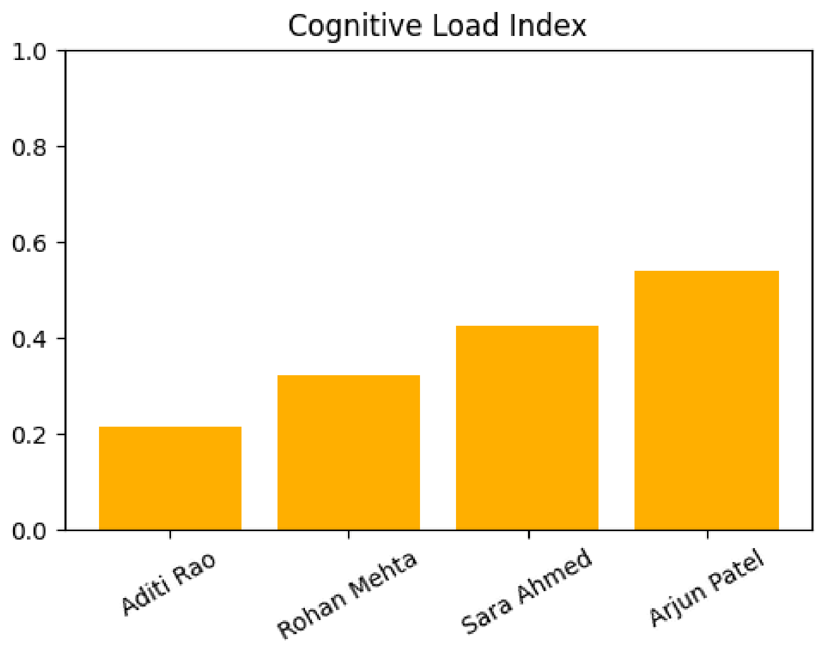
Step 5: Decision Logic for HR Task Assignment Recommendation

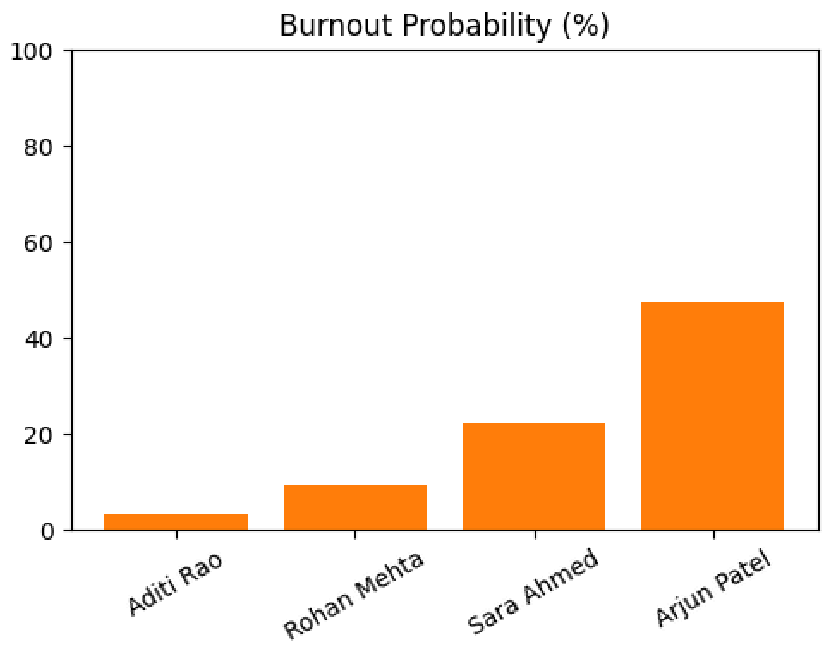
Based on CLI, Burnout Probability, ΔPerformance, and Cluster Persona, the system recommends:

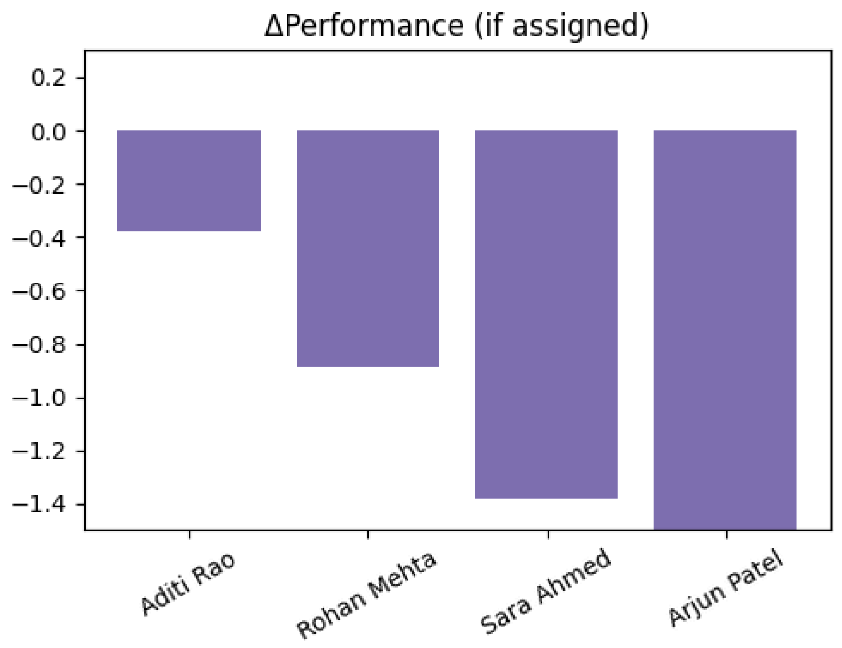
- 🟢 Assign Normally //low load, stable behavior
- ⚠️ Assign With Support //moderate strain, manageable risk
- 🚫 Do Not Assign //high load, high stress, or severe ΔPerformance drop
- 🔥 High Priority Assignment //minimal performance impact and good stability

This creates an end-to-end ML-driven workload management tool capable of supporting real HR decisions.

Result’s of Integrated Simulation System







Employee: Aditi Rao
Role: Mid
Department: Engineering
CLI: 0.217
Burnout Probability: 3.5%
ΔPerformance: -0.376
Cluster: 3 (Healthy & Balanced)
Risk: Low
HR Action: 🟢 Assign Normally

Employee: Rohan Mehta
Role: Senior
Department: DataScience
CLI: 0.323
Burnout Probability: 9.4%
ΔPerformance: -0.889
Cluster: 1 (Stable & Low Risk)
Risk: Medium
HR Action: ⚠️ Assign With Support

Employee: Sara Ahmed
Role: Lead
Department: Product
CLI: 0.425
Burnout Probability: 22.3%
ΔPerformance: -1.378
Cluster: 2 (High Work & Stress)
Risk: Medium
HR Action: 🚫 Do Not Assign

Employee: Arjun Patel
Role: C-Level
Department: Operations
CLI: 0.541
Burnout Probability: 47.8%
ΔPerformance: -1.792
Cluster: 2 (High Work & Stress)
Risk: Medium
HR Action: 🚫 Do Not Assign